In a Different Voice

Gender Composition and Scientific Writing in Economics

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Hengel (2022) documented a readability differential; this paper extends the analysis

Hengel (2022)

Female-authored papers are **more readable**.

The differential *widens during peer review* —
consistent with higher writing standards.

One dimension measured:
Flesch Reading Ease.

→

This paper

Does the differential extend to other writing dimensions?

We extend the analysis to **15 additional**
surface-level writing dimensions:
directness, jargon, novelty-claim strength,
assertiveness, hedging, evidentiary support,
emotional valence, and more.

9,117 abstracts. 16-item LLM rubric.

9,117 abstracts scored on 16 surface features by GPT-5

9,117
abstracts

AER ECA JPE QJE
1950–2015

$<10\%$ female authors overall

$\sim 3\%$ all-female articles

$\sim 9\%$ majority-female

16 criteria scored 1–10, surface features only:

G1 Creativity & Hedging

G2 Assertiveness & Voice

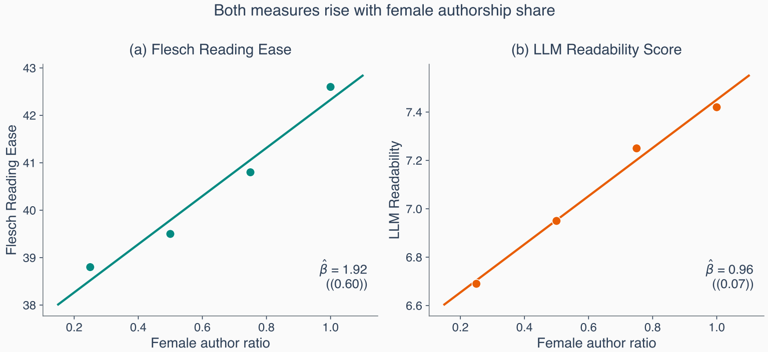
G3 Structural Directness

G4 Authorial Stance & Novelty

G5 Support & Impact

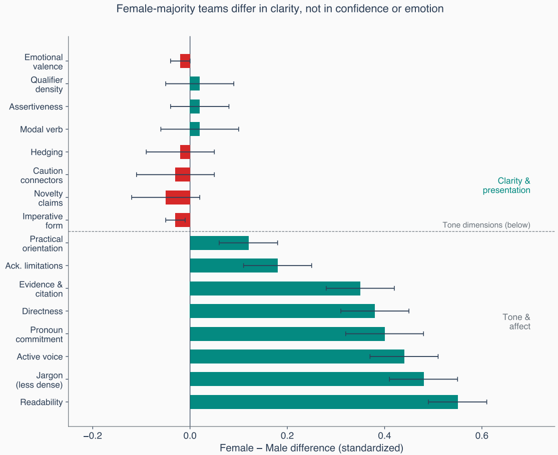
Standalone: Readability

LLM readability replicates Hengel's established pattern — $\hat{\beta} = 0.36^{***}$



LLM readability coefficient **0.36** ($p < 0.01$) is **significant across all nine specifications**. Hengel's Flesch: 1.25^{**}.

Female-majority teams score systematically higher on clarity; tonal differences are negligible

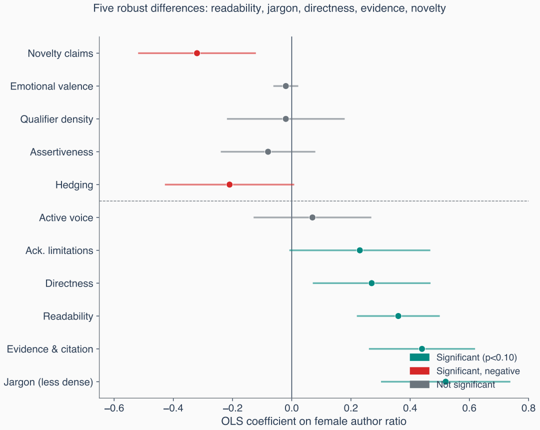


Five clarity dimensions are systematically elevated; tonal dimensions yield null differences

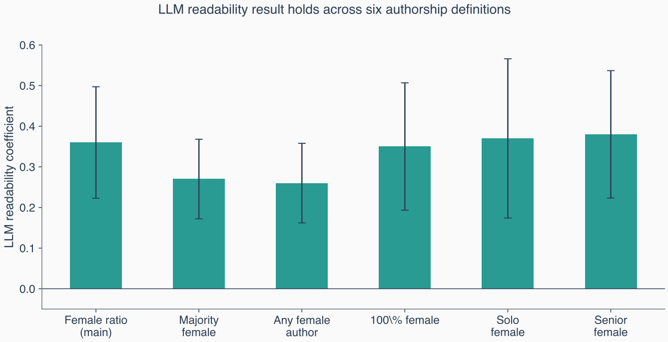
Female-Majority Higher		Statistically Negligible
Readability	$+0.55$ SD	Assertiveness
Jargon density (lower)	$+0.48$ SD	Hedging frequency
Active/passive voice	$+0.44$ SD	Qualifier density
Sentence directness	$+0.38$ SD	Modal verb strength
Evidentiary support	$+0.35$ SD	Emotional valence
Novelty-claim strength	-0.05 SD	Caution-signaling connectors

Unconditional; majority-gender classification; all criteria standardized.

Controlled regressions confirm five robust differentials; tonal dimensions yield null estimates



The readability differential is robust across six authorship specifications



OLS coefficient on female authorship indicator, Column 1 specification. Error bars denote 95% confidence intervals. Results for jargon, directness, and evidentiary support are similarly robust.

Identification concern: GPT-5 may introduce systematic measurement bias

Potential confound

GPT-5 may assign systematically

higher scores to abstracts by female-majority teams, independent of underlying linguistic content.

Response

(1) LLM readability replicates Hengel's

manually calculated scores.

(2) Results are consistent across six authorship specifications.

(3) A model exhibiting such bias would produce elevated scores on assertive-ness

and hedging. The data show no such pattern.

Female-majority teams in economics write more clearly.

Not more hesitantly. Not more emotionally.

The difference lies in how research is *communicated*,

not in the confidence with which findings are asserted.

Backup: Empirical specification

$$R_j = \beta_0 + \beta_1 \cdot \mathit{FemaleRatio}_j + \gamma X_j + \delta Y_j + \varepsilon_j$$

X_j: journal×year FE, editor controls, blind-review indicator, JEL codes, theory/empirical flag

Y_j: institution FE, author prominence (max T FE), author seniority (max t FE), native-speaker indicator

Nine specifications. SEs clustered on editor.
Alternative authorship measures: majority-female binary, any-female binary, 100% female, solo, senior female.

Backup: Summary statistics by gender composition

	All-Male ($N = 7{,}951$)	Mixed ($N = 853$)	All-Female ($N = 313$)	Full ($N = 9{,}117$)
Readability	6.70 (1.38)	7.42 (1.25)	7.37 (1.30)	6.79 (1.38)
Directness	5.94 (2.08)	6.71 (1.81)	6.53 (1.86)	6.03 (2.06)
Jargon (raw)	7.00 (1.73)	6.48 (1.86)	6.07 (1.92)	6.92 (1.76)
Evidence	1.68 (1.34)	2.31 (1.90)	2.19 (1.73)	1.75 (1.43)
Novelty	2.27 (2.12)	2.26 (2.20)	1.90 (1.85)	2.25 (2.12)
Assertiveness	8.18 (1.55)	8.43 (1.28)	8.04 (1.55)	8.20 (1.53)
Hedging	8.35 (1.98)	8.49 (1.73)	8.07 (2.11)	8.35 (1.96)
Emotional val.	1.12 (0.37)	1.11 (0.35)	1.10 (0.31)	1.12 (0.37)

Means and SDs. Scored 1–10. Descriptive only; no significance stars.

Backup: Key regression coefficients (col. 1, full sample)

Criterion	Coef.	SE
Female > male (significant)		
Jargon (less dense)	$+0.52^{***}$	(0.11)
Evidence & citation	$+0.44^{***}$	(0.09)
Readability	$+0.36^{***}$	(0.07)
Directness	$+0.27^{**}$	(0.10)
Female < male (significant)		
Novelty claims	-0.32^{***}	(0.10)
Not significant		
Assertiveness	-0.08	(0.08)
Emotional valence	-0.02	(0.02)

OLS, female ratio, 1950–2015. SEs clustered on editor. $N = 9{,}117$.